

## 7.4 Operational Properties II

In this section, we will introduce more useful properties of the Laplace Transform.

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$$\text{Let } F(s) = \mathcal{L}\{f(t)\}.$$

By definition

$$F(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

Then:

$$F'(s) = \frac{d}{ds} \int_0^{\infty} e^{-st} f(t) dt$$

You can bring the derivative inside the integral as a partial derivative.

$$= \int_0^{\infty} \frac{\partial}{\partial s} (e^{-st} f(t)) dt$$

$$F'(s) = - \int_0^{\infty} e^{-st} t f(t) dt$$

$$\Rightarrow F'(s) = - \mathcal{L}\{t f(t)\}$$

$$\Rightarrow \boxed{\mathcal{L}\{t f(t)\} = - F'(s)}$$

More generally,

$$\boxed{\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)}$$

for  $n = 1, 2, 3, \dots$

$F^{(n)}(s)$  is the  $n^{\text{th}}$  derivative of  $F(s)$ .

Suppose that  $F(s) = \mathcal{L}\{\sin(\omega t)\}$ .

$$F(s) = \frac{\omega}{s^2 + \omega^2} = \omega(s^2 + \omega^2)^{-1}$$

$$\begin{aligned}\Rightarrow F'(s) &= -\omega(s^2 + \omega^2)^{-2} (2s) \\ &= \frac{-2\omega s}{(s^2 + \omega^2)^2}\end{aligned}$$

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Then:  $\mathcal{L}\{t \sin(\omega t)\} = -F'(s)$

$$\Rightarrow \boxed{\mathcal{L}\{t \sin(\omega t)\} = \frac{2\omega s}{(s^2 + \omega^2)^2}}$$

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Similarly, we can derive that:

$$\boxed{\mathcal{L}\{t \cos(\omega t)\} = \frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}}$$

ex: ① Find  $\mathcal{L}\{t e^{3t} \sin(4t)\}$ .

Method ①

$$\mathcal{L}\{t f(t)\} = -F'(s)$$

$$\mathcal{L}\{e^{3t} \sin(4t)\} = \frac{4}{(s-3)^2 + 16} = F(s)$$

$$\Rightarrow \mathcal{L}\{t e^{3t} \sin(4t)\} = -F'(s)$$

$$= \frac{8(s-3)}{[(s-3)^2 + 16]^2}$$

Method ②

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$$

$$\mathcal{L}\{t \sin(4t)\} = \frac{8s}{(s^2 + 16)^2}$$

$$\Rightarrow \mathcal{L}\{t e^{3t} \sin(4t)\} = \frac{8(s-3)}{[(s-3)^2 + 16]^2}$$

ex: ② Evaluate  $\mathcal{L}^{-1} \left\{ \frac{s+2}{(s^2+4s+13)^2} \right\}$ .

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$s^2+4s+13$  is irreducible.

Complete the square

$$\begin{aligned} s^2+4s+13 &= (s^2+4s+4) + 9 \\ &= (s+2)^2 + 9 \end{aligned}$$

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Recall that:

$$\boxed{\mathcal{L}^{-1} \{ F(s-a) \} = e^{at} \mathcal{L}^{-1} \{ F(s) \}}$$

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$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{s+2}{[(s+2)^2+9]^2} \right\} = e^{-2t} \mathcal{L}^{-1} \left\{ \frac{s}{(s^2+9)^2} \right\}$$

$$\mathcal{L}\{t \sin(\omega t)\} = \frac{2\omega s}{(s^2 + \omega^2)^2}$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{s}{(s^2 + \omega^2)^2}\right\} = \frac{t \sin(\omega t)}{2\omega}$$


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Therefore :

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s^2+4s+13)^2}\right\} = e^{-2t} \mathcal{L}^{-1}\left\{\frac{s}{(s^2+9)^2}\right\}$$

$$= e^{-2t} \left( \frac{t \sin(3t)}{2(3)} \right)$$

$$= \boxed{\frac{t e^{-2t} \sin(3t)}{6}}$$

ex: (3) Solve the IVP

$$\begin{cases} x''(t) + 4x(t) = \cos(2t) \\ x(0) = 1 \\ x'(0) = 3 \end{cases}$$

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$$\mathcal{L}\{x''(t)\} + 4\mathcal{L}\{x(t)\} = \mathcal{L}\{\cos(2t)\}$$

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Let  $X(s) = \mathcal{L}\{x(t)\}$ ,

$$\Rightarrow [s^2 X(s) - sx(0) - x'(0)] + 4X(s) = \frac{s}{s^2 + 4}$$

$$\Rightarrow (s^2 + 4)X(s) - s - 3 = \frac{s}{s^2 + 4}$$

$$\Rightarrow X(s) = \frac{s + 3}{s^2 + 4} + \frac{s}{(s^2 + 4)^2}$$

$$X(s) = \frac{s}{s^2+4} + \frac{3}{s^2+4} + \frac{s}{(s^2+4)^2}$$

$$X(s) = \frac{s}{s^2+4} + \frac{3}{2} \left( \frac{2}{s^2+4} \right) + \frac{1}{4} \left( \frac{4s}{(s^2+4)^2} \right)$$

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$$x(t) = \mathcal{L}^{-1} \{ X(s) \}$$

$$x(t) = \cos(2t) + \frac{3}{2} \sin(2t) + \frac{1}{4} t \sin(2t)$$

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ex: ④ Solve the following IVP.

$$\begin{cases} y''(t) + 2y'(t) + 5y(t) = 3e^{-t}\cos(2t) \\ y(0) = 1 \\ y'(0) = 2 \end{cases}$$

Take Laplace transform on both sides of the DE to get:

$$\left[ s^2 Y(s) - sy(0) - y'(0) \right] + 2 \left[ sY(s) - y(0) \right] + 5Y(s) = \frac{3(s+1)}{(s+1)^2 + 4}$$

$$(s^2 + 2s + 5)Y(s) - s - 2 - 2 = \frac{3(s+1)}{(s+1)^2 + 4}$$

$s^2 + 2s + 5$  is irreducible

Complete the square:  $s^2 + 2s + 5 = (s+1)^2 + 4$

$$\Rightarrow Y(s) = \frac{s+4}{(s+1)^2 + 4} + \frac{3(s+1)}{[(s+1)^2 + 4]^2}$$

$$Y(s) = \frac{(s+1) + 3}{(s+1)^2 + 4} + \frac{3(s+1)}{[(s+1)^2 + 4]^2}$$

We will use the property:

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at} \mathcal{L}^{-1}\{F(s)\}$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

$$y(t) = e^{-t} \mathcal{L}^{-1}\left\{\frac{s+3}{s^2+4} + \frac{3s}{(s^2+4)^2}\right\}$$

$$y(t) = e^{-t} \mathcal{L}^{-1}\left\{\frac{s}{s^2+4} + \frac{3}{2}\left(\frac{2}{s^2+4}\right) + \frac{3}{4}\left(\frac{4s}{(s^2+4)^2}\right)\right\}$$

$$y(t) = e^{-t} \cos(2t) + \frac{3}{2} e^{-t} \sin(2t) + \frac{3}{4} t e^{-t} \sin(2t)$$

# Convolution

The convolution of  $f(t)$  and  $g(t)$  is another function of  $t$  defined by

$$f(t) * g(t) = \int_0^t f(\theta) g(t-\theta) d\theta$$

↓  
The convolution of  
 $f(t)$  and  $g(t)$

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The order does not matter:

$$f(t) * g(t) = g(t) * f(t)$$

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Note: the textbook uses  $\tau$  instead of  $\theta$  for the variable of integration.

## Convolution theorem

$$\text{If } \begin{cases} F(s) = \mathcal{L}\{f(t)\} \text{ and} \\ G(s) = \mathcal{L}\{g(t)\}, \end{cases}$$

then

$$\mathcal{L}\{f(t) * g(t)\} = F(s) G(s).$$

It then follows that:

$$\mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t)$$

ex: ⑤ Use convolution to evaluate

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s+3)(s-4)} \right\}.$$

$$F(s) = \frac{1}{s+3} \xrightarrow{\mathcal{L}^{-1}} f(t) = e^{-3t}$$

$$G(s) = \frac{1}{s-4} \xrightarrow{\mathcal{L}^{-1}} g(t) = e^{4t}$$

$$\mathcal{L}^{-1} \{ F(s) G(s) \} = f(t) * g(t)$$

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{1}{(s+3)(s-4)} \right\} &= e^{-3t} * e^{4t} \\ &= \int_0^t e^{-3\theta} e^{4(t-\theta)} d\theta \end{aligned}$$

$$= \int_0^t e^{4t} e^{-3\theta} e^{-4\theta} d\theta$$

$$= e^{4t} \int_0^t e^{-7\theta} d\theta$$

$$= e^{4t} \left( \frac{e^{-7\theta}}{-7} \right) \bigg|_{\theta=0}^{\theta=t}$$

$$= -\frac{e^{4t}}{7} (e^{-7t} - 1)$$

$$= \boxed{-\frac{1}{7} e^{-3t} + \frac{1}{7} e^{4t}}$$

Note The purpose of example (5) was to illustrate how convolution works. It would have been faster to use P.F. as follows.

$$\frac{1}{(s+3)(s-4)} = \frac{A}{s+3} + \frac{B}{s-4}$$

$$\Rightarrow \begin{cases} A = -1/7 \\ B = 1/7 \end{cases}$$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{1}{(s+3)(s-4)} \right\} = -\frac{1}{7} e^{-3t} + \frac{1}{7} e^{4t}$$

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The next example could not be solved with P.F., we will use Convolution.

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ex: ⑥ Evaluate  $\mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)^2} \right\}$ .

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$$F(s) = \frac{1}{s^2+1} \xrightarrow{\mathcal{L}^{-1}} f(t) = \sin t$$

$$G(s) = \frac{1}{s^2+1} \xrightarrow{\mathcal{L}^{-1}} g(t) = \sin t$$

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$$\frac{1}{(s^2+1)^2} = F(s)G(s)$$

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$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)^2} \right\} = f(t) * g(t)$$

$$= \sin t * \sin t$$

$$= \int_0^t \sin \theta \sin(t-\theta) d\theta$$

We will use the identity :

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

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$$\begin{aligned}\sin t * \sin t &= \int_0^t \sin \theta \sin(t-\theta) d\theta \\ &= \frac{1}{2} \int_0^t [\cos(2\theta-t) - \cos t] d\theta \\ &= \frac{1}{2} \left[ \frac{\sin(2\theta-t)}{2} - \theta \cos t \right]_{\theta=0}^{\theta=t}\end{aligned}$$

$$= \frac{1}{2} \left[ \left( \frac{\sin t}{2} - t \cos t \right) - \left( \frac{\sin(-t)}{2} - 0 \right) \right]$$

$$= \frac{1}{2} \left( \frac{\sin t}{2} - t \cos t - \frac{\sin(-t)}{2} \right)$$

$$\sin(-t) = -\sin t$$

$$\Rightarrow \sin t * \sin t = \frac{1}{2} (\sin t - t \cos t)$$

Then:

$$\boxed{\mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)^2} \right\} = \frac{\sin t - t \cos t}{2}}$$

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More generally,

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s^2+\omega^2)^2} \right\} = \frac{\sin(\omega t) - \omega t \cos(\omega t)}{2 \omega^3}$$

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The above formula appears in our table of Laplace Transforms.

$$\mathcal{L}\{f(t) * g(t)\} = F(s) G(s)$$

$$\text{Where: } f(t) * g(t) = \int_0^t f(\theta) g(t-\theta) d\theta$$

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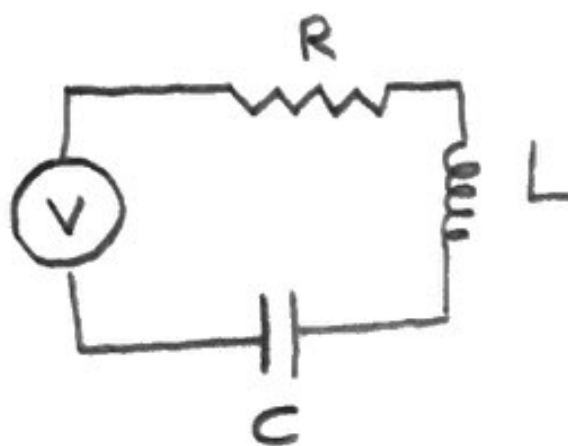
An important special case is

$$\text{when } g(t) = 1 \rightarrow G(s) = \frac{1}{s}$$

$$f(t) * 1 = \int_0^t f(\theta) d\theta$$

$$\Rightarrow \boxed{\mathcal{L}\left\{\int_0^t f(\theta) d\theta\right\} = \frac{F(s)}{s}}$$

## LRC - circuits



$$\begin{cases} q(t) = \text{charge} \\ i(t) = \frac{dq}{dt} = \text{current} \end{cases}$$

As we saw in section 5.1,  
The DE for the charge  $q(t)$  is

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q(t) = V(t)$$

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Since  $i(t) = \frac{dq}{dt}$ , then:

$$q(t) = q(0) + \int_0^t i(\theta) d\theta$$

Therefore,

$$L \frac{di}{dt} + R i(t) + \frac{1}{C} \left( q(0) + \int_0^t i(\theta) d\theta \right) = V(t)$$

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ex: ⑦ Consider an LRC-circuit

with:  $L = \frac{5}{3} \text{ H}$ ,  $R = 10 \Omega$

$C = \frac{1}{30} \text{ F}$ ,  $V = 300 \text{ V}$

and initial conditions:  $q(0) = 0$  and

$$i(0) = 0.$$

Find the current  $i(t)$ .

$$L \frac{di}{dt} + R i(t) + \frac{1}{C} \left( q(0) + \int_0^t i(\theta) d\theta \right) = V(t)$$

$$\Rightarrow \frac{5}{3} \frac{di}{dt} + 10 i(t) + 30 \int_0^t i(\theta) d\theta = 300$$

$$\Rightarrow 5 \frac{di}{dt} + 30 i(t) + 90 \int_0^t i(\theta) d\theta = 900$$

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Take Laplace Transform.

$$\Rightarrow 5 \mathcal{L} \left\{ \frac{di}{dt} \right\} + 30 \mathcal{L} \{ i(t) \} + 90 \mathcal{L} \left\{ \int_0^t i(\theta) d\theta \right\} \\ \dots = \mathcal{L} \{ 900 \}$$

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$$5 \left[ s I(s) - i(0) \right] + 30 I(s) + \frac{90 I(s)}{s} = \frac{900}{s}$$

$$\Rightarrow (5s^2 + 30s + 90) I(s) = 900$$

$$\Rightarrow (s^2 + 6s + 18) I(s) = 180$$

$$\Rightarrow I(s) = \frac{180}{s^2 + 6s + 18}$$

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$s^2 + 6s + 18$  is irreducible.

Complete the square

$$\begin{aligned} s^2 + 6s + 18 &= (s^2 + 6s + 9) + 9 \\ &= (s+3)^2 + 9 \end{aligned}$$

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$$I(s) = \frac{180}{(s+3)^2 + 9}$$

$$= 60 \left( \frac{3}{(s+3)^2 + 9} \right)$$

$$i(t) = \mathcal{L}^{-1} \{ I(s) \}$$

$$i(t) = 60 e^{-3t} \sin(3t)$$

